Lecture plan · for the lecturer

Linear programming and duality

Optimization · University · Lecture 1

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Formulate LPs.
- See why corners are optimal.
- Interpret duals.

Activity: Draw it, then price it

1. Warm-up (5 min): Give the canteen's targets and ask students to guess the cheapest menu in two minutes.
2. Draw (20 min): Pairs shade the feasible region and evaluate every vertex.
3. Dual (15 min): Write the dual and check both optimal values match.
4. Price (15 min): Explain each shadow price to the canteen manager in one sentence.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which corner won?
- What does a zero shadow price mean?
- When does a price stop being valid?

Exit ticket

- Explain one shadow price in plain words.

Try it in your next lecture
Give pairs a two-variable canteen problem on graph paper; they shade the feasible region, test every corner and find the optimum, then interpret the solver’s shadow prices for the manager.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Shadow price 5, extra hour costs 3. Buy it?**

◯ Yes ◯ Shadow prices say nothing about this ◯ No

2 **Solve at the corners.**

max 3x + 2y s.t. x + y ≤ 4, x + 3y ≤ 6, x, y ≥ 0. The optimum is _____ at x = _____.

Word bank: 11 · 4 · 12 · 3

3 **Which statements about LP duality are true? Choose all that apply.**

☐ The dual has one variable per primal constraint

☐ At optimality, primal and dual values are equal

☐ A non-binding constraint has a positive shadow price

☐ Any feasible dual value bounds the primal optimum

4 **A shadow price is valid only within its constraint's sensitivity range.**

◯ True ◯ False

5 **Match each term to its meaning.**

Vertex	• • • •	Dual values bound the primal
Weak duality	• • • •	Where an optimum can be found
Complementary slackness	• • • •	Value of one more unit of a resource
Shadow price	• • • •	Slack constraint means zero dual

Answers · for the lecturer

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1 **Yes**

Each extra hour improves the objective by 5 and costs 3: a net gain of 2, while the price stays valid.

2 **max 3x + 2y s.t. x + y ≤ 4, x + 3y ≤ 6, x, y ≥ 0. The optimum is 12 at x = 4.**

Vertices (0,0), (4,0), (3,1), (0,2) give 0, 12, 11, 4.

3 **Any feasible dual value bounds the primal optimum; At optimality, primal and dual values are equal; The dual has one variable per primal constraint**

By complementary slackness, slack constraints have zero shadow prices.

4 **True**

It holds only within the sensitivity range, until the optimal basis changes.

5 **Vertex — Where an optimum can be found**
Weak duality — Dual values bound the primal
Complementary slackness — Slack constraint means zero dual
Shadow price — Value of one more unit of a resource

Duality connects geometry to economics.

For you to think about

Do your students spend more time pivoting tableaux or interpreting solutions? What would you rebalance?